Register Number: 192425119

Question 3: Average Traffic Density

Research Question

Is the average traffic density significantly different from 50 units?

Hypotheses

- H0: $\mu = 50$ units
- H1: $\mu \neq 50$ units (two-sided test)
- Significance level $\alpha = 0.05$

Formula

$t = (\bar{x} - \mu_o) / (s / \sqrt{n})$

$df = n - 1$

Data Used (read from reference figure)

Quantity	Value
Sample size n	100
Sample mean $\bar{x}$	51.3
Sample SD s (estimated from histogram)	≈ 10.3
Hypothesised mean μ_o	50

Calculation

- $t = (51.3 - 50) / (10.3 / \sqrt{100}) = 1.3 / 1.03 \approx 1.26$
- df = 99, two-sided p-value from the t distribution.

Result

Quantity	Value
Sample mean	51.3
t statistic	1.26
df	99
Two-sided p-value	0.21

Conclusion

The p-value (0.21) is greater than 0.05, so we fail to reject H0. The average traffic density is not significantly different from 50 units.

Interpretation Notes

- The sample standard deviation was estimated from the histogram bin midpoints, so the t statistic and p-value are approximate. The histogram is slightly right-skewed, which is acceptable for n = 100 under the t-test.
- The sample mean (51.3) is close to the hypothesised value of 50, so a small difference is expected to be non-significant.

Graph

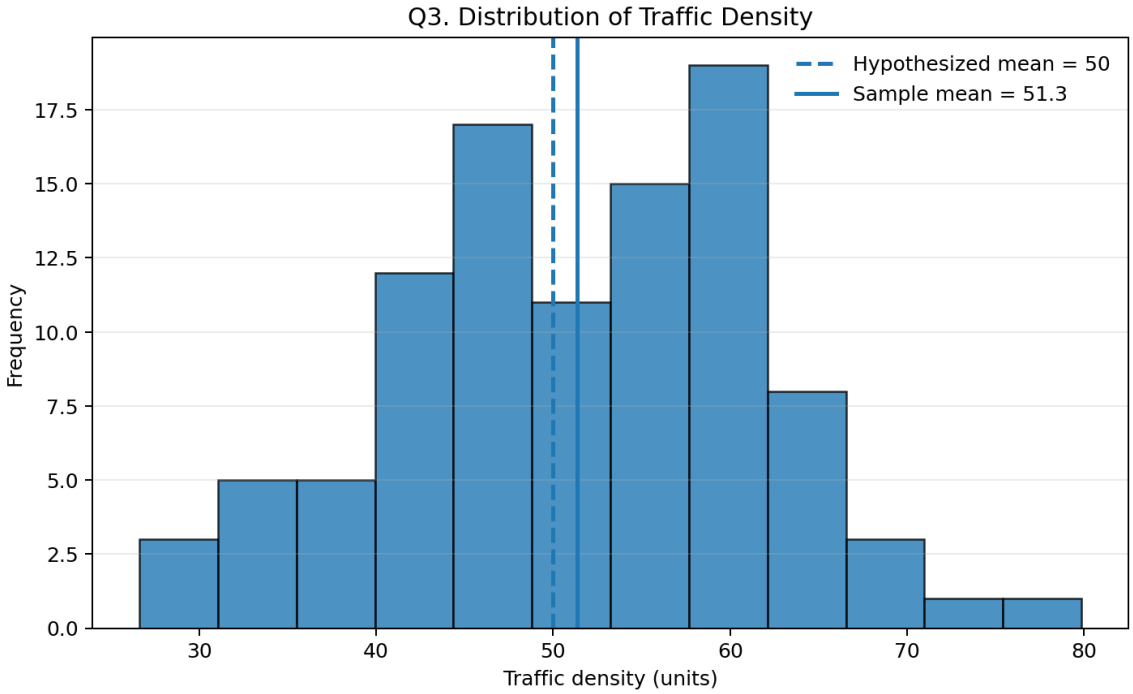


Figure 3. Reference graph for Question 3.

Note: the counts and values in this document were read from the reference figure. Re-run the test on the original project dataset in R (e.g. chisq.test, t.test, aov, cor.test) to confirm the final numbers before submission.